

National Aeronautics and Space Administration

SARASOTA CLIMATE

Monitoring Heat and Assessing Heat Vulnerability to Identify Locations for Heat Mitigation Efforts in Sarasota, Florida

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DEVELOP

Georgia – Athens | Spring 2024

Study Area

Sarasota County, Florida

- **Study Period**: Summer 2019-2023 (June to August)



430,000+ residents



14.4% population growth



40% aged 65 & older



250+ days of sunshine



Humid Subtropical climate



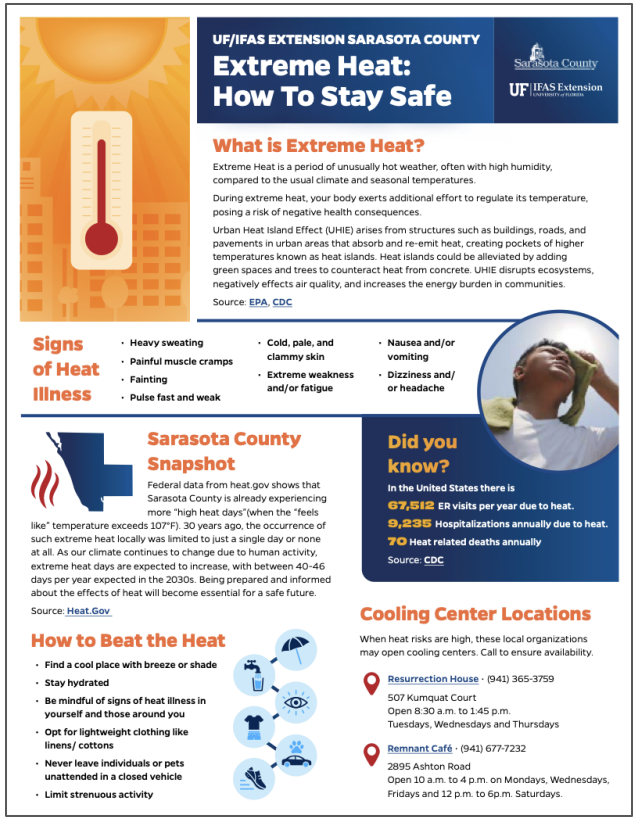
Icon credits: Flaticon, Freepik

Partner

Sarasota County Univ. of Florida's Institute of Food and Agricultural Sciences extension

A collaborative program between Sarasota County, the University of Florida, and the US Department of Agriculture. The program integrates expert research into community activities across 6 major areas:

- 4-H youth development
- Agriculture
- Gardening and landscaping
- Natural resources
- Nutrition and healthy living
- **Sustainability**



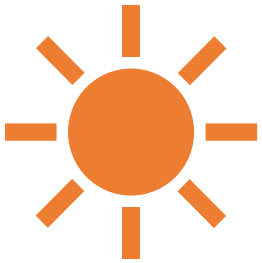
Sarasota County Extreme Heat Outreach Poster

Image credit: Sarasota County UF/IFAS Extension

The sustainability group is interested in identifying areas to kick-start **extreme heat mitigation efforts**, such as **tree planting and building cooling centers.**

Extreme Heat in Sarasota County

Since the 1970s, Sarasota has faced a significant rise in temperatures during the summer season.



+2.6°F



+5.9°F

Current Average Summer Temperature: 85°F

2023 was recorded as the warmest year with:

- **63 consecutive hot days (heat index HI > 100°F)**
- **1 full week in August of HI ≥ 110°F**

With climate change impacts becoming more evident,
the number of hot days might continually increase to **131 days per year around 2050!***

*Union of Concerned Scientists (2019), mid-century 2036-2065 estimates

Extreme Heat Impacts

Cities have started becoming heat traps due to the fast expansion of urban development. This phenomenon happens as a result of heat being absorbed and retained by buildings, roads, and other infrastructures which end up creating ***"heat islands"*** with temperatures noticeably higher than that in rural counterparts.

Several problems emerged from prolonged exposure of extreme heat:

Negative impacts on **air quality** and exacerbation of **respiratory conditions**

Increased numbers of **heat-related illness**

Higher **energy consumption** and associated costs for cooling buildings

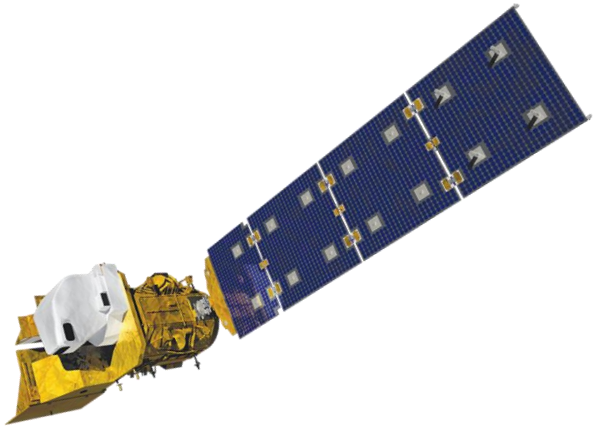
Disproportionate heat exposure and **risks in low-income** and underserved communities

Objectives

1. Assess land surface temperature variance across Sarasota County
2. Explore the relationship between vulnerable populations and severe heat
3. Visualize heat anomalies, heat mitigation capacity, and heat vulnerability to support community outreach

Earth Observations



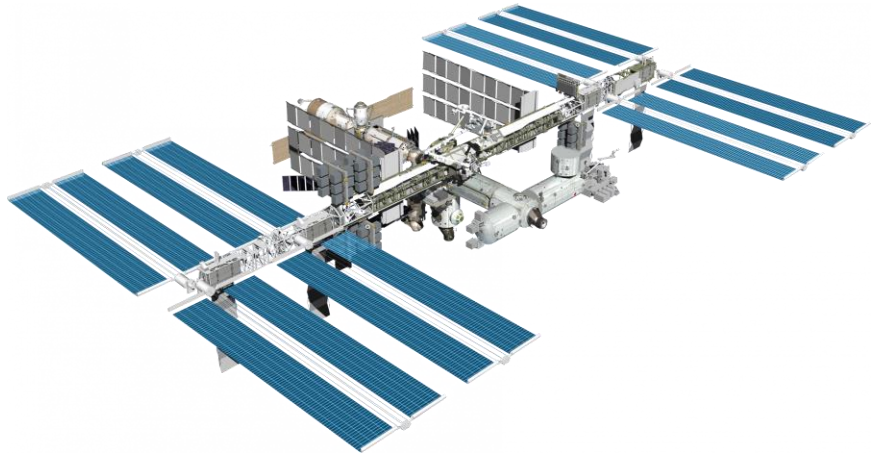


Landsat 8 Landsat 9

Operational Land Imager & Thermal Infrared Sensors
OLI - TIRS

Spatial resolution: 30 meters
Temporal revisitation: 16 days

Daytime LST, Albedo, Provisional Evapotranspiration



ISS

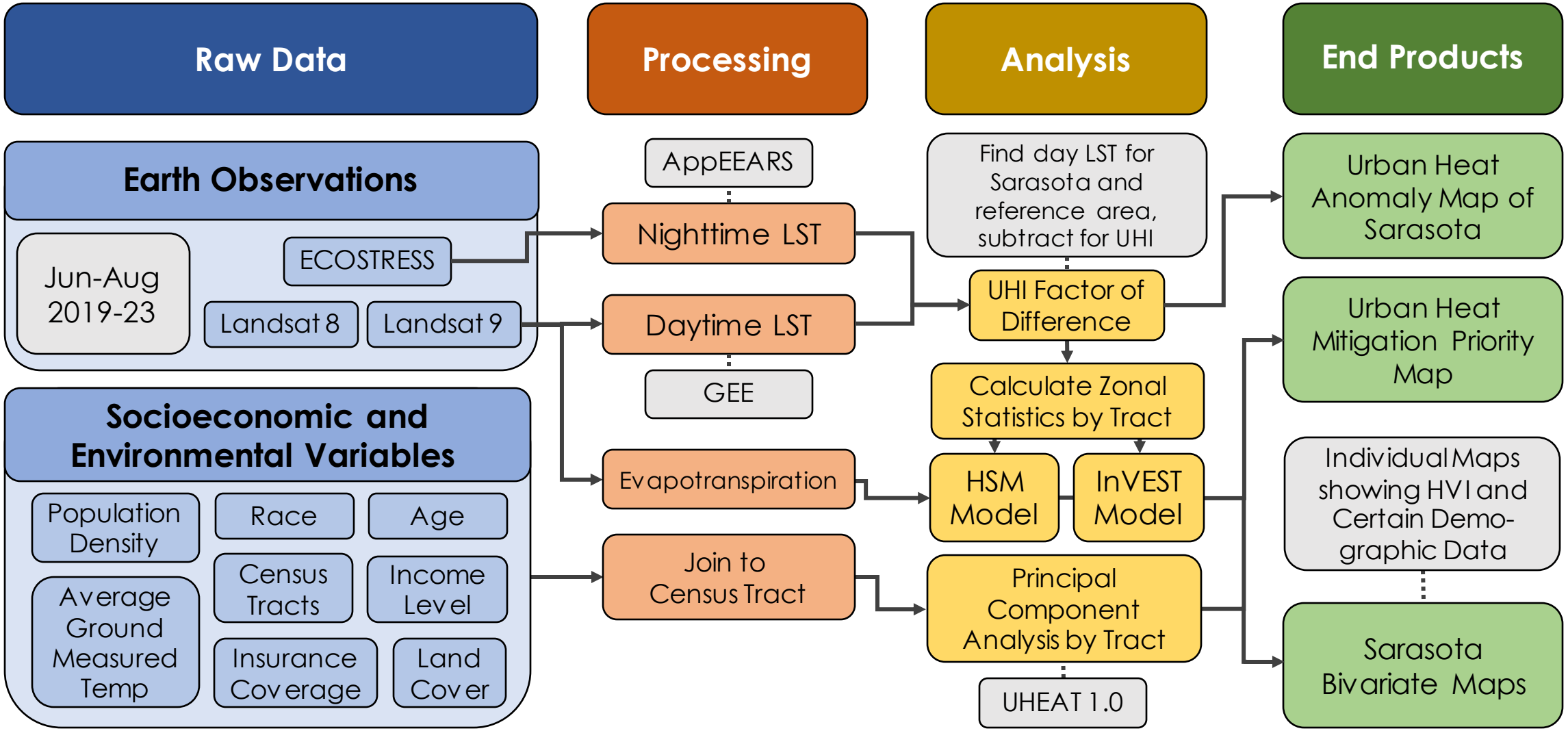
ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station
ECOSTRESS

Spatial resolution: 70 meters
Temporal revisitation: 1-7 days

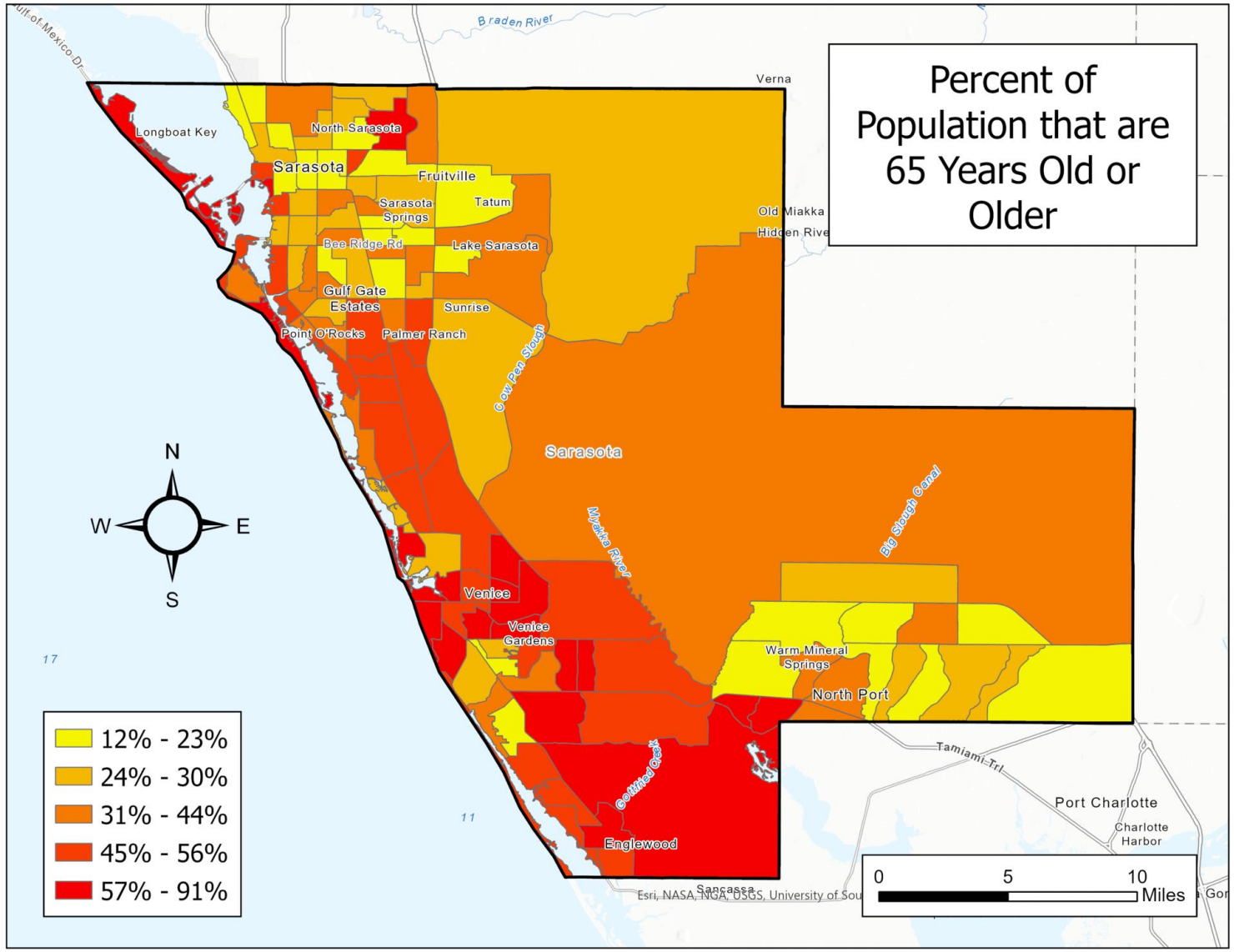
Nighttime LST

Image credit: NASA

Methodology



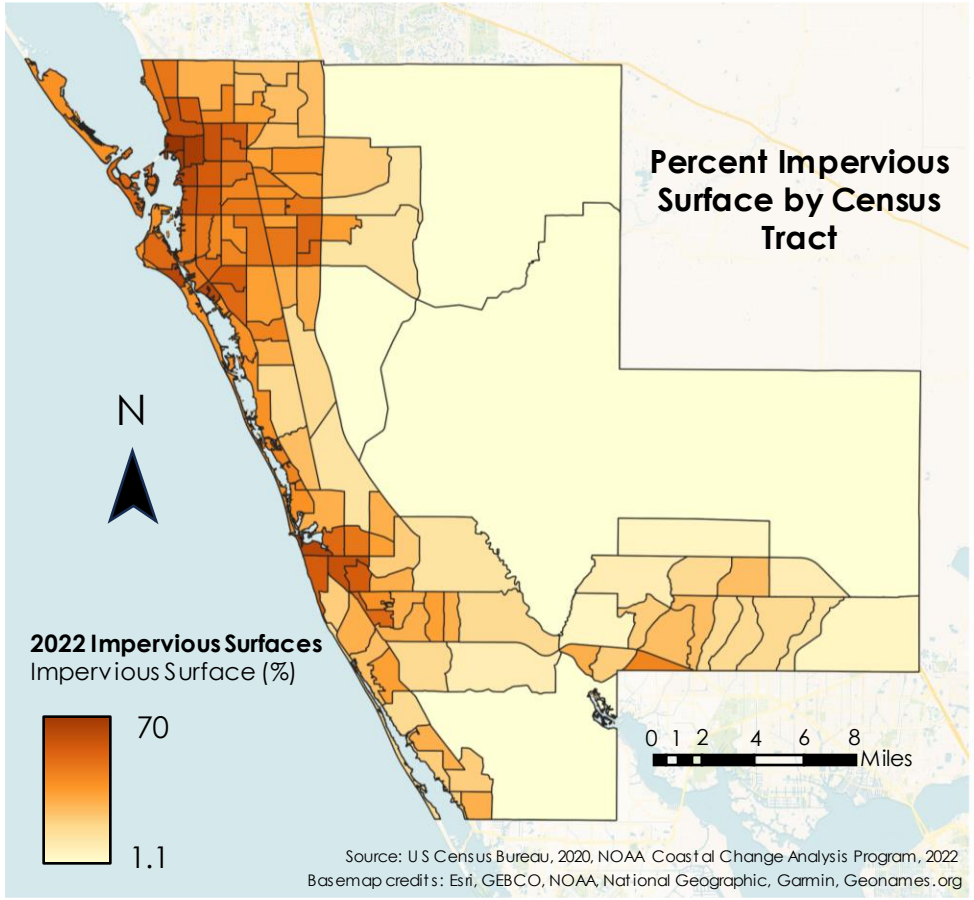
Demographic Data



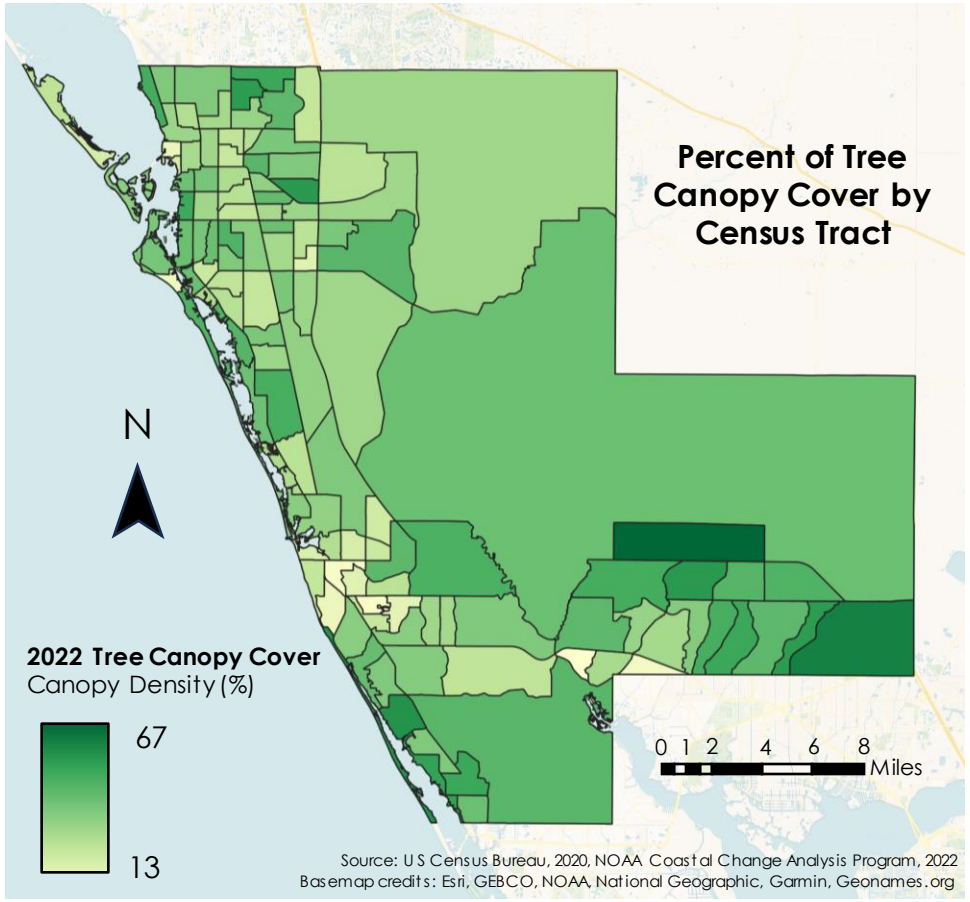
Basemap credits: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

- Highest **population density** around **Sarasota City**, including a spike in North Sarasota, as well as the cities of North Port and Venice
- Demographic risks (often) **associated with age** are clustered largely in the **southern part** of the county, while most **other variables** are centered mostly in **the northeast**
- **North Sarasota and North Port both reflect high percentages in almost every category**
  - Poverty, Living Alone, Population with No Car, Unemployed, etc.
  - Both types of risks present

Land Cover



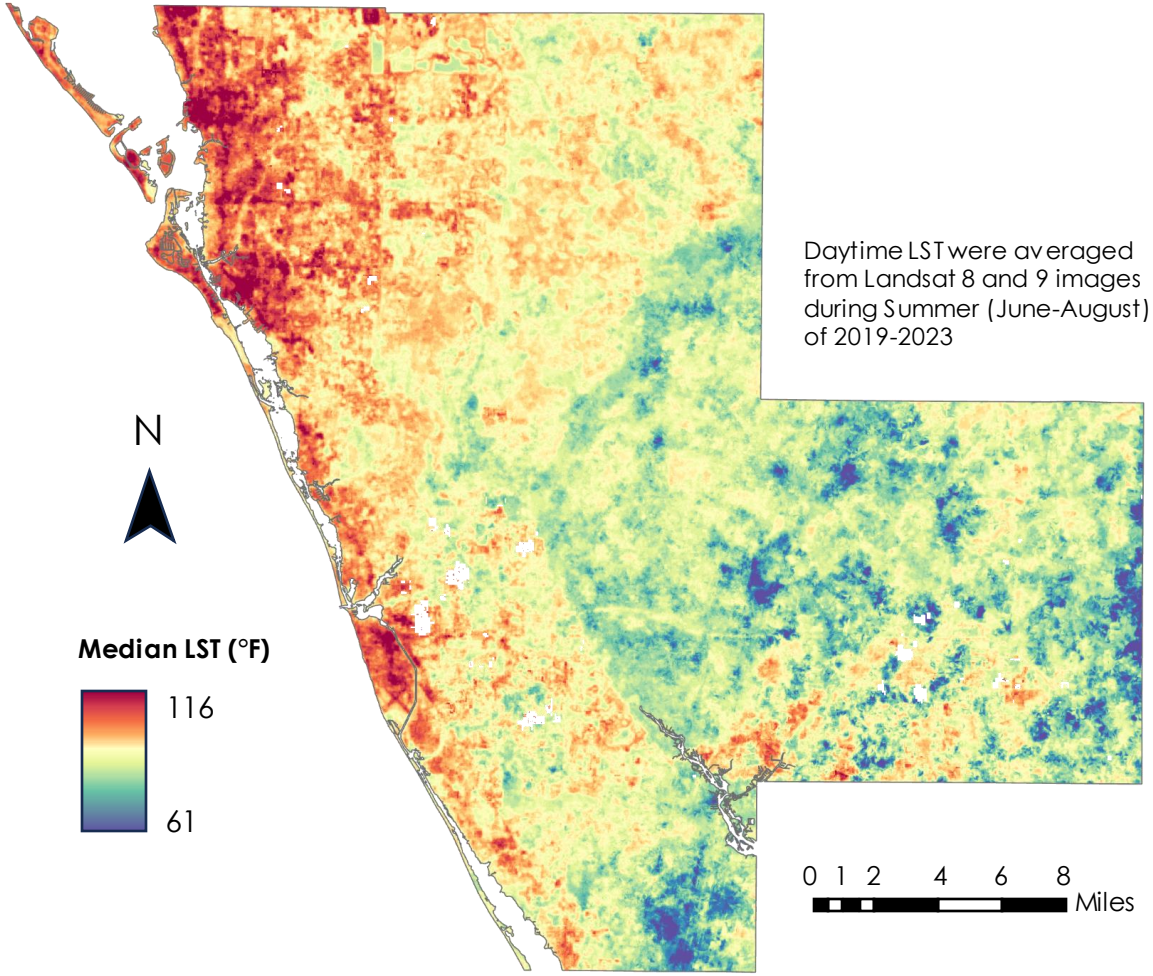
These two maps were derived from 2022 NOAA Coastal Change Program dataset with resolution 1m x 1m



- The **western side** of the county is where most **urban areas** are located, which was shown from the higher density of impervious surfaces.
- While the **eastern side** is where the **state parks and pastures** are located, depicted by the average ~ 35% of canopy cover.

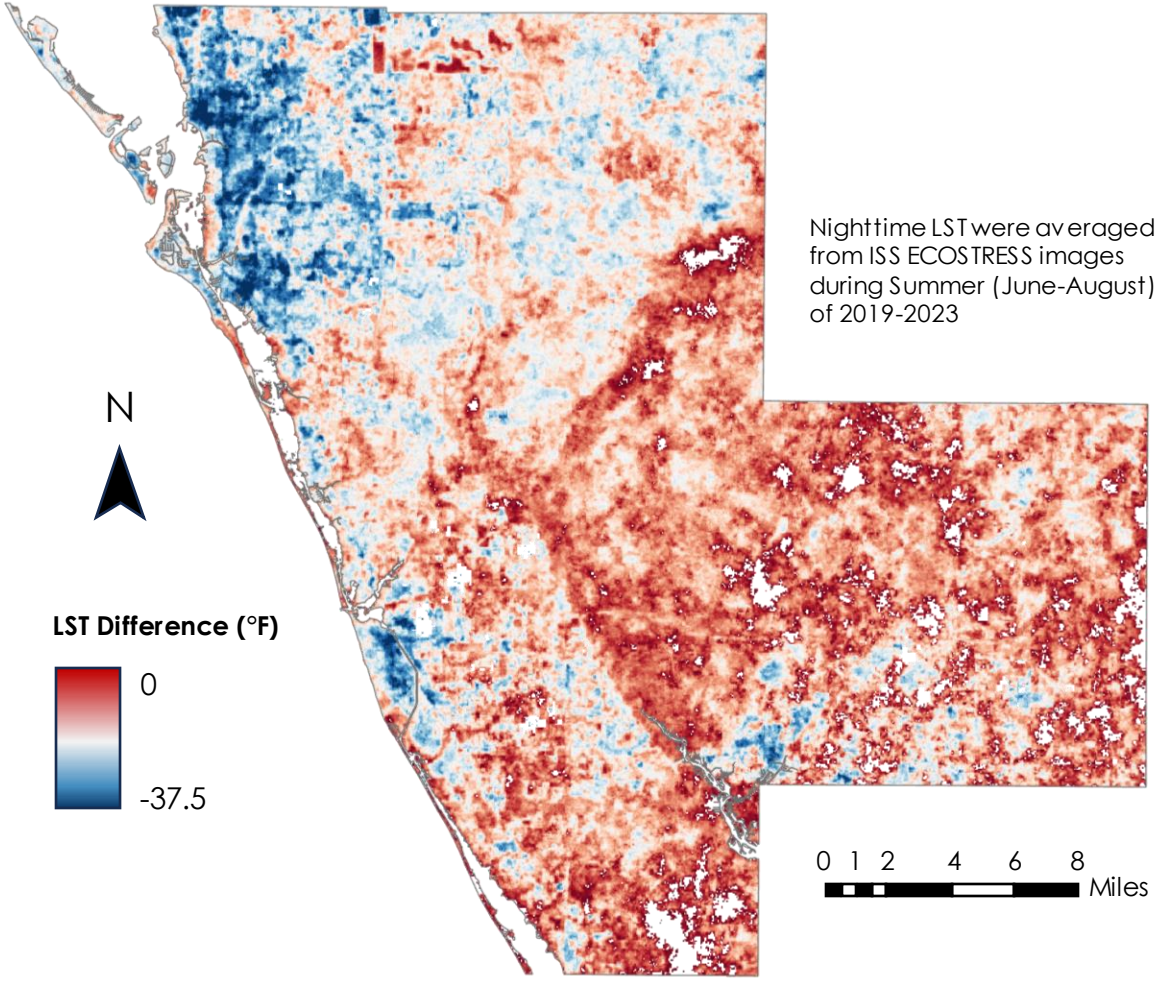
Land Surface Temperature

Daytime



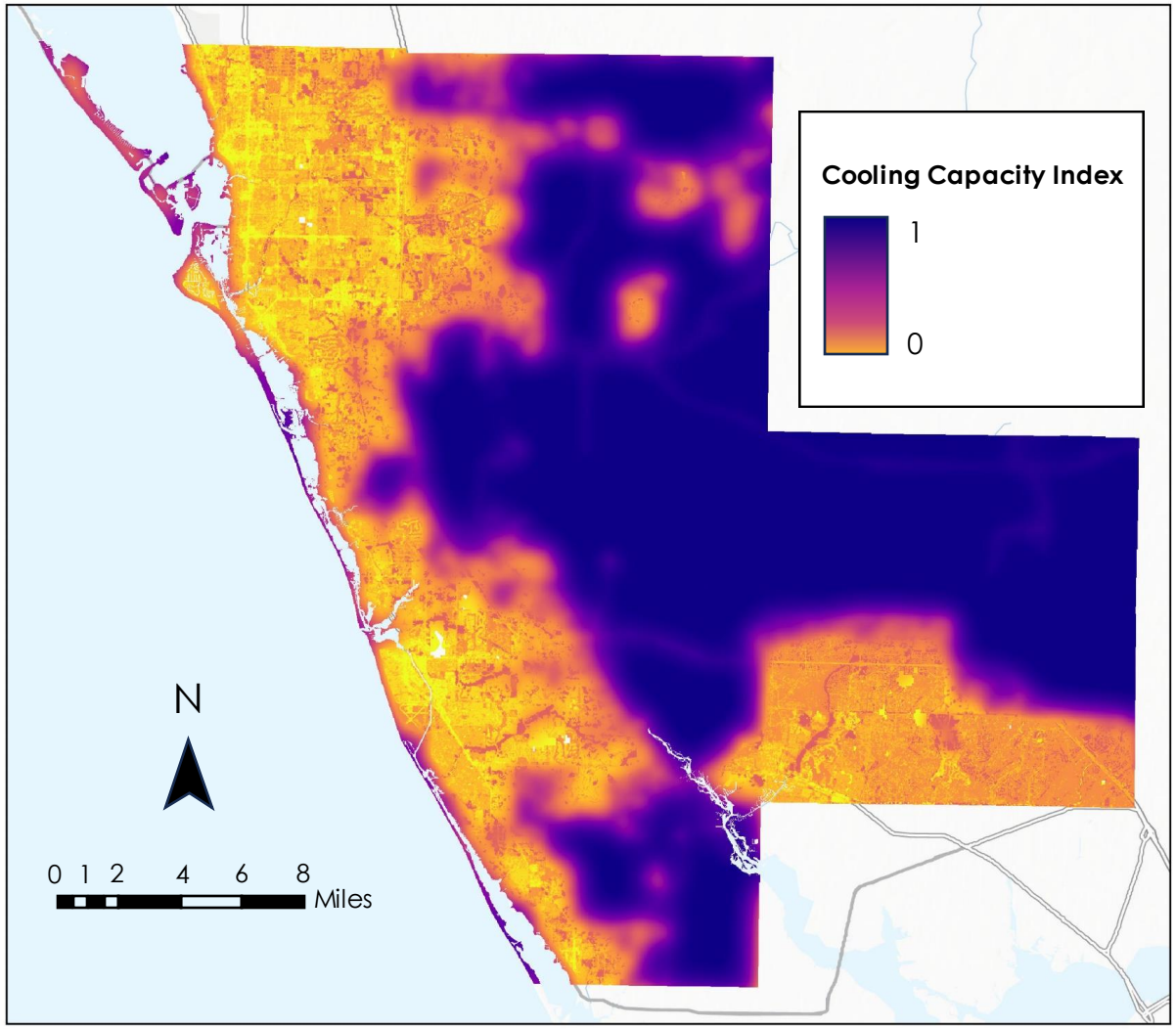
Daytime LST were averaged from Landsat 8 and 9 images during Summer (June-August) of 2019-2023

Nighttime Cooling



Nighttime LST were averaged from ISS ECOSTRESS images during Summer (June-August) of 2019-2023

Cooling Capacity Index

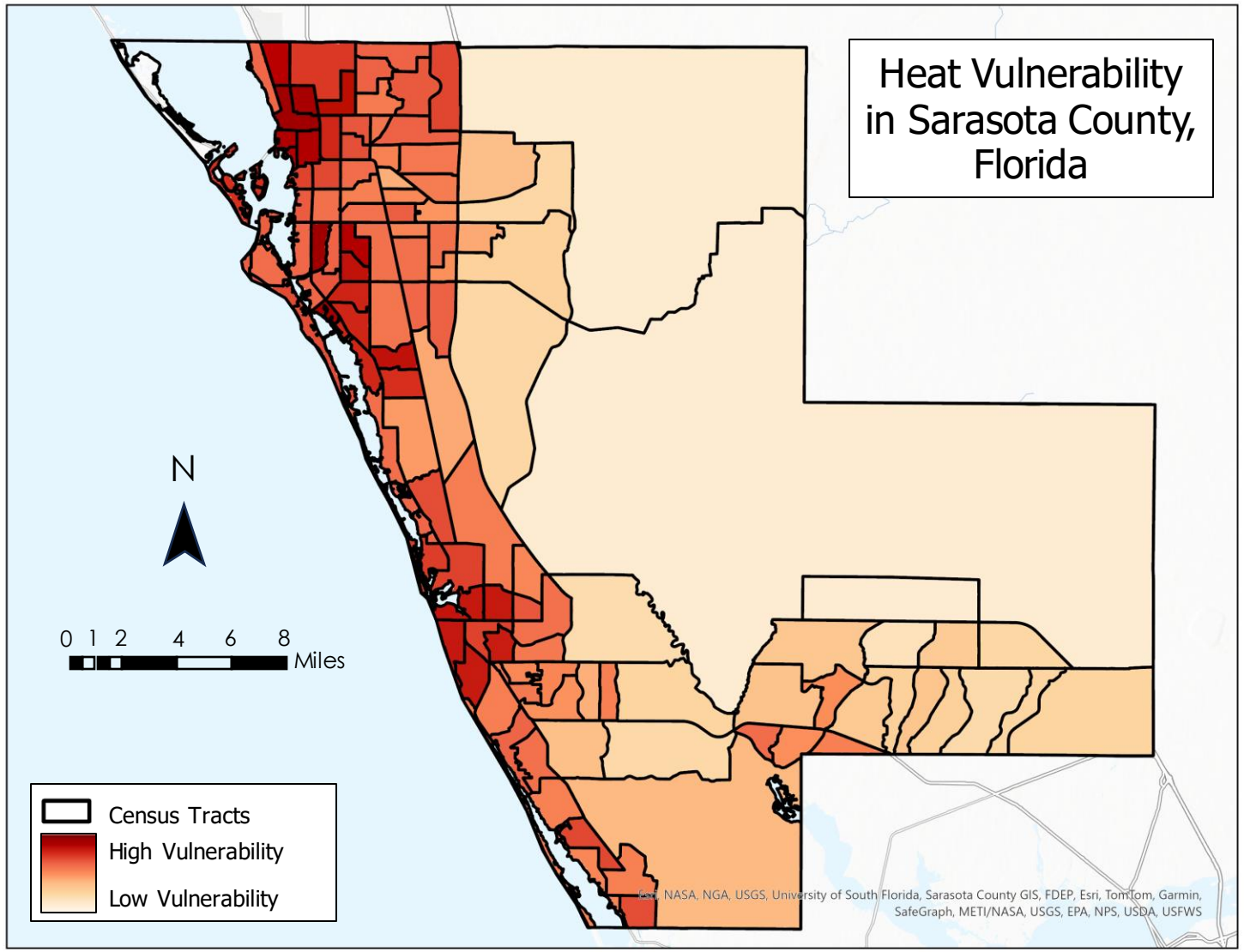


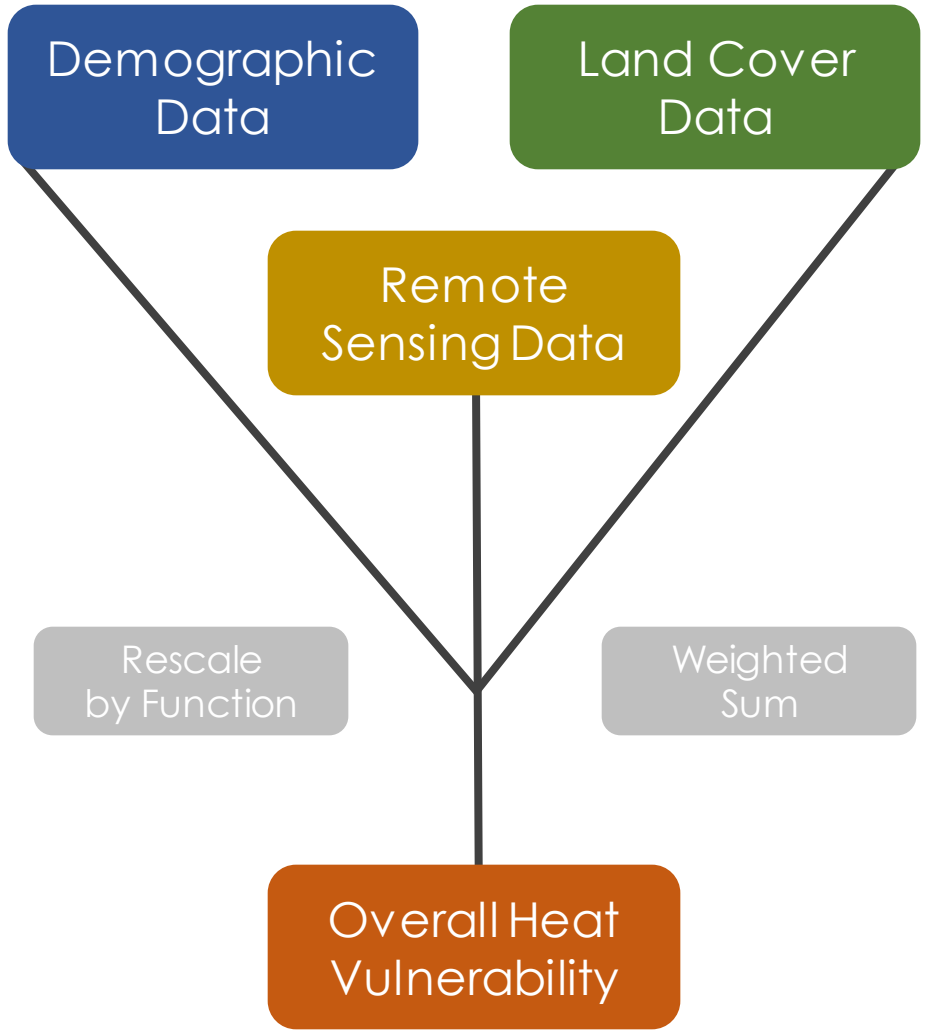
Basemap credits: Esri, NASA, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Ability to dissipate heat based on:

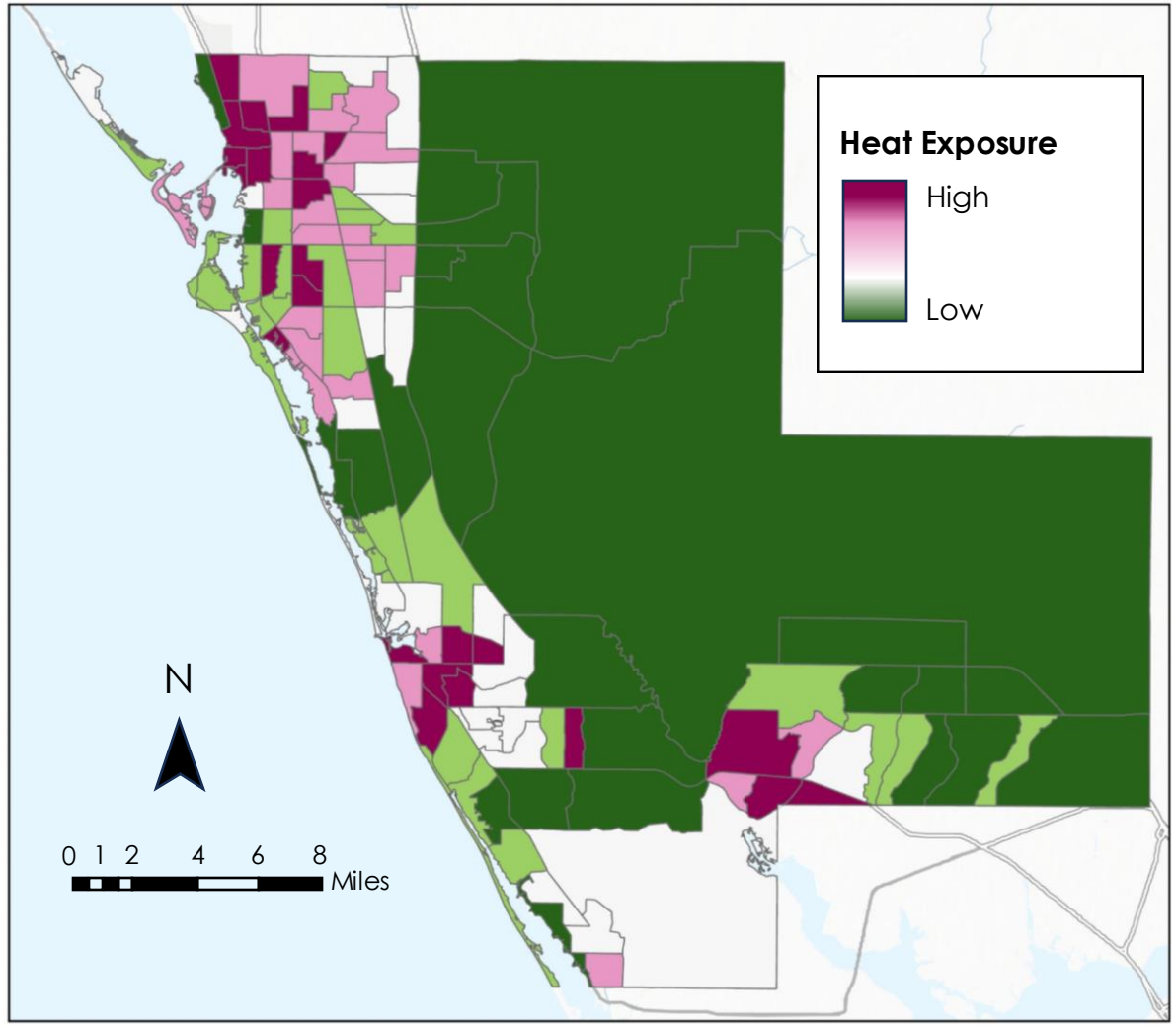
- Shade
- Evapotranspiration
- Biophysical properties
- Distance from cooling islands

Heat Vulnerability (from Lit. Review)





Heat Exposure (PCA Analysis)



Basemap credits: Esri, NASA, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Map Credits: Theresia Phoa

Environmental (5 Variables):
Daytime LST, Nighttime LST, Albedo, % Canopy Cover, % Impervious Surfaces

Socioeconomic (11 Variables):
Total Population, Age, Race, Disability Status, Income Level, Health Insurance Ownership, Car Ownership, English Proficiency, Occupational Status, % Living Alone

Total: 16 Variables

PCA Analysis:
Reduce multidimensionality and show most relevant variables

4 Principal Components

(See appendix slide)

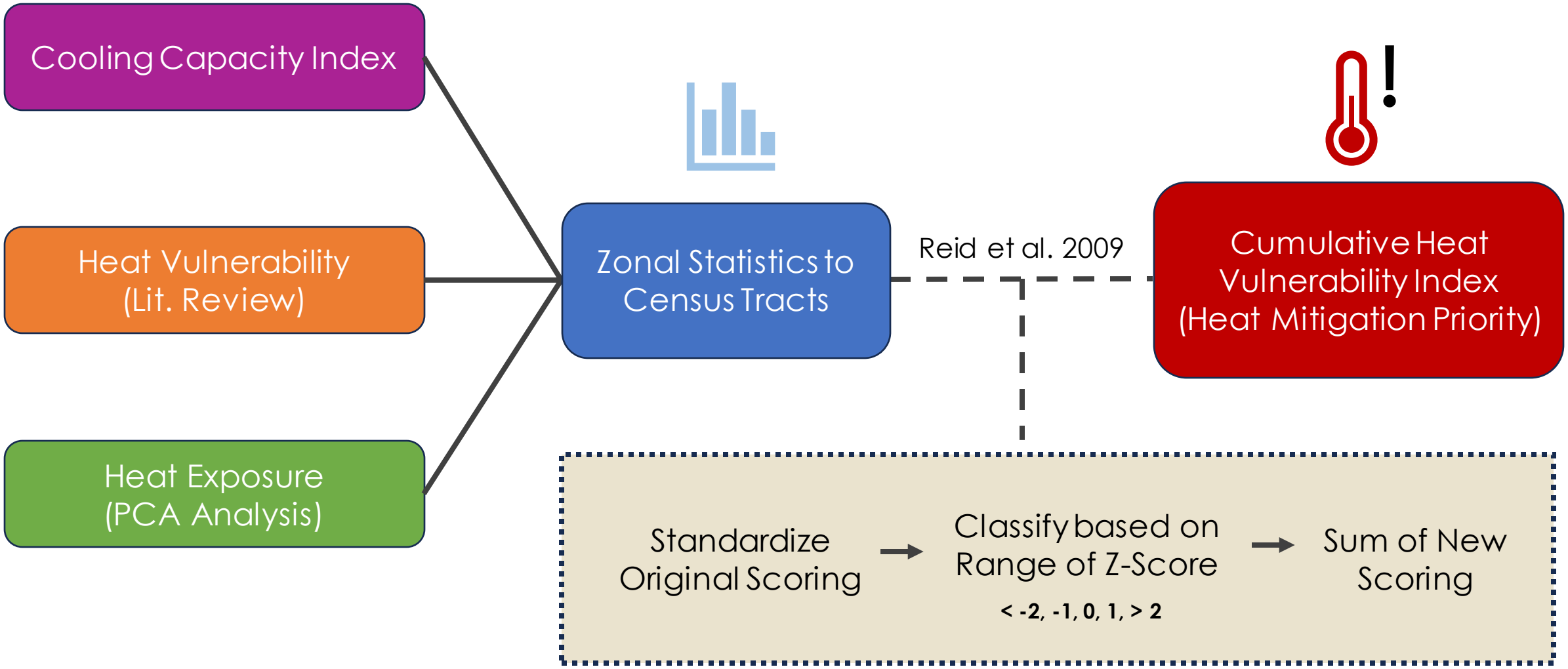
Land Cover
Albedo
Population

Daytime LST

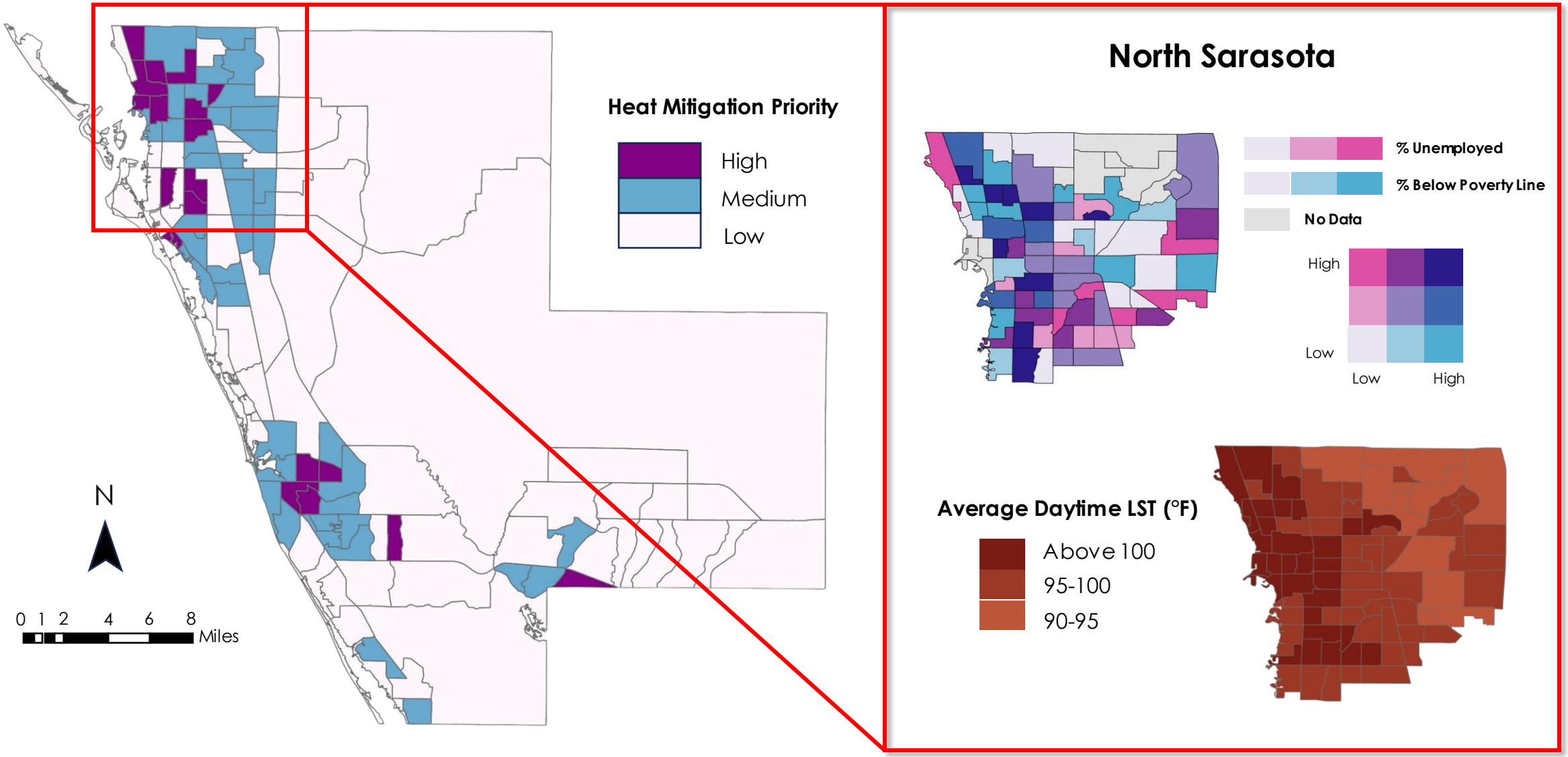
Age (65 and Over)
Occupational Status

Nighttime LST

Determining Priority Areas



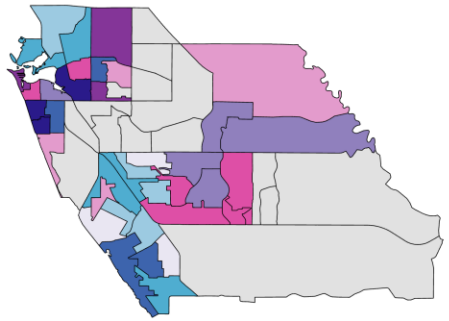
Heat Mitigation Priority (1/3)



Map Credits: Catherine Gallagher and Theresia Phoa

Heat Mitigation Priority (2/3)

Venice



% Age 65 and over

% Below Poverty Line

No Data

High

Low

Low High

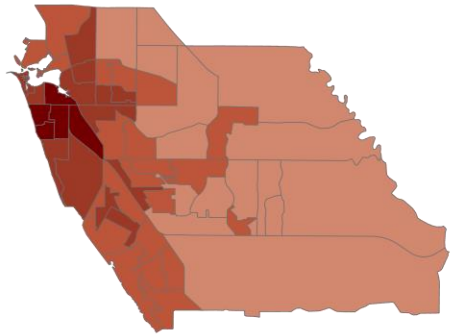
Average Daytime LST (°F)

Above 100

95-100

90-95

85-90



Heat Mitigation Priority

High

Medium

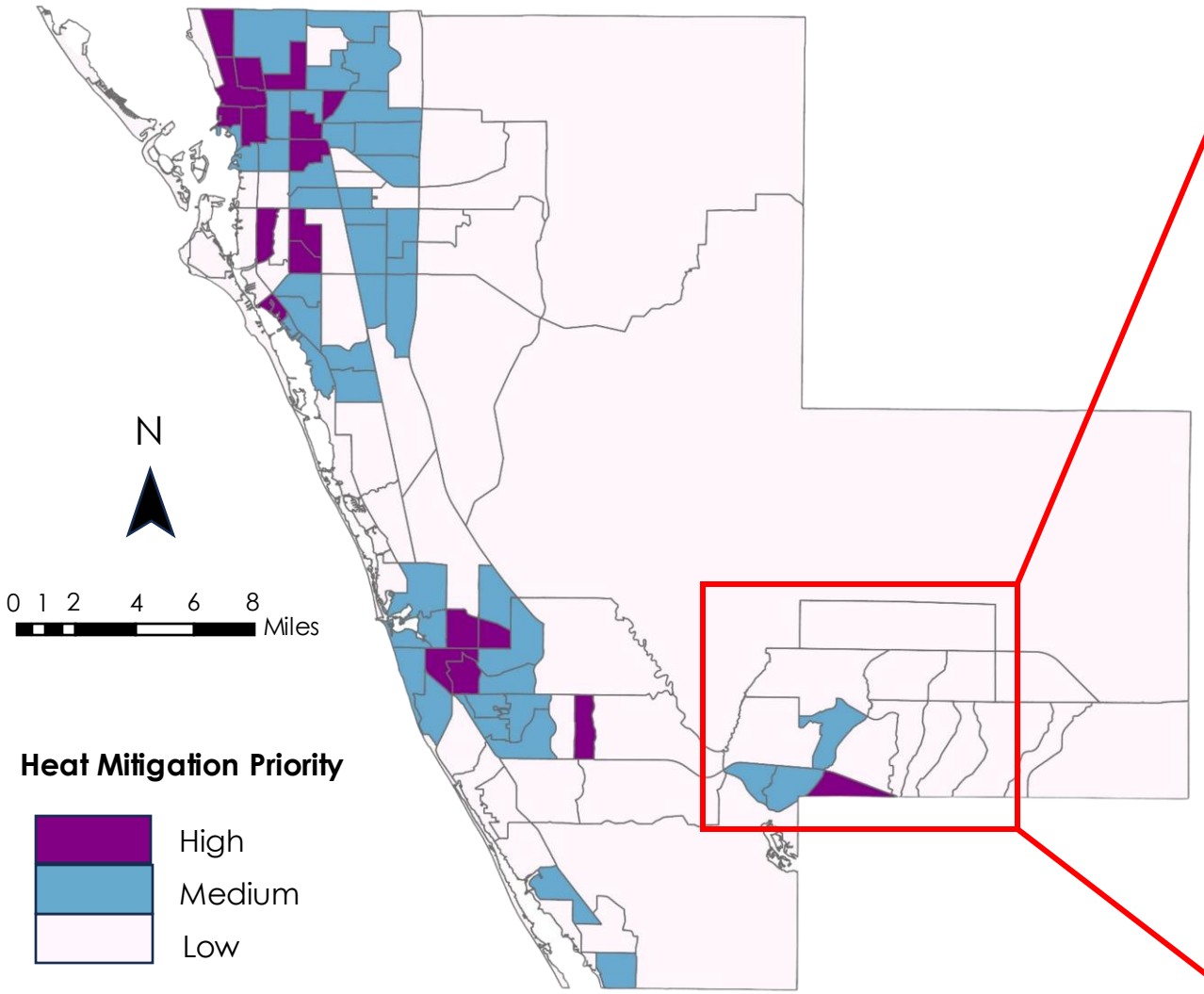
Low

N

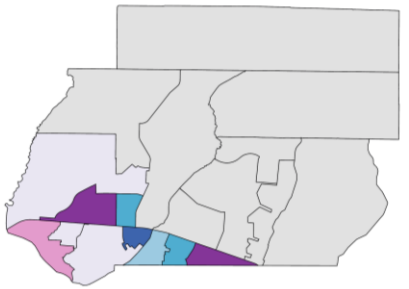
0 1 2 4 6 8

Miles

Heat Mitigation Priority (3/3)



North Port



% Unemployed

% Below Poverty Line

No Data

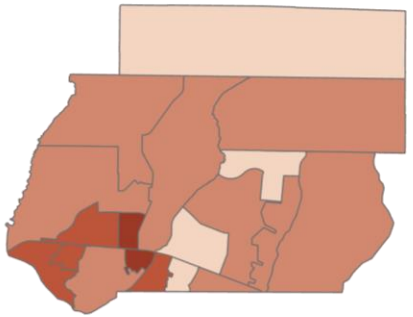
High

Low

Low High

Average Daytime LST (°F)

95-100
90-95
85-90
80-85



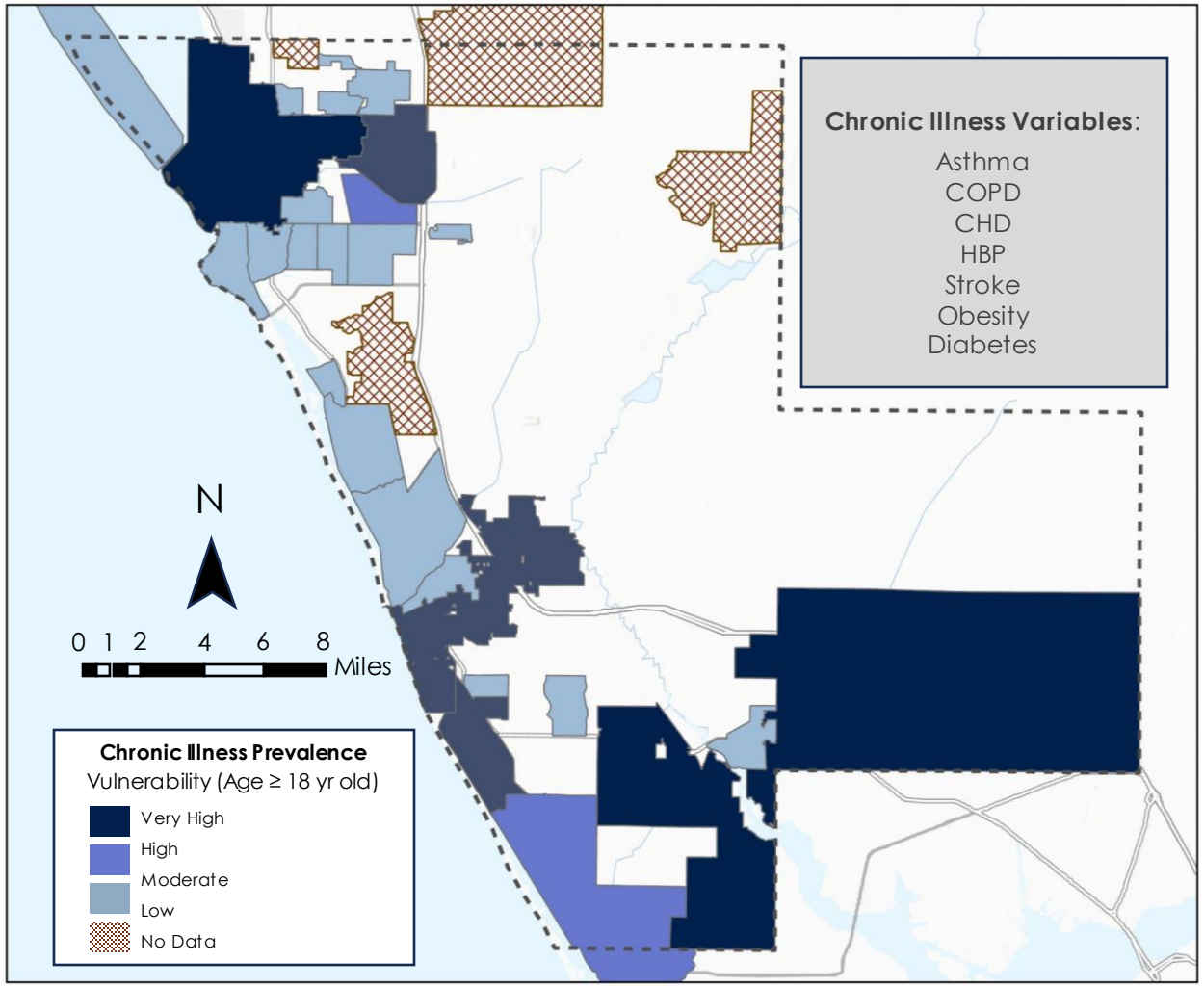
Conclusions

- Heat intensity **varies significantly** across Sarasota County, with the **highest temperatures** in the **developed western portion** of the county.

- Many socio-economically vulnerable populations live in high-heat areas, particularly in **North Sarasota, Venice, and North Port.**

- Tree canopy cover has a **generally negative correlation** with heat intensity, which could provide a **strategy for mitigation** in vulnerable areas.

Limitations – Health Data



Basemap credits: Esri, NASA, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

- 2020 BRFSS Data (CDC Telephone Surveys)
- Data only available by **Census Designated Places** (equivalent to City, Townships, Village)
- Age-adjusted percentage **(Adults ≥ 18 yr old)**
- **Aggregated** using Cumulative Index method (Reid et al. 2009) **by number of cases of Chronic Illnesses**:
  - Asthma
  - Chronic obstructive pulmonary disease (COPD)
  - Coronary heart disease (CHD)
  - High Blood Pressure (HBP)
  - Stroke
  - Obesity
  - Diabetes

Future Recommendations

- Future work should investigate potential mitigation methods to address extreme heat, such as:
  - Create an interactive web map of heat island data that will be available to the public
  - Monitor priority areas using low-cost temperature sensors
  - Increasing tree canopy cover in urban heat islands
  - Placing cooling centers in vulnerable communities
  - Increasing albedo by installing light-colored roofs

Partner

- Further studies should include smaller scale regions using high-resolution microclimate models (Envi-MET, SOLWEIG, etc.)

DEVELOP

Acknowledgments

- **Partners:** Alia Garrett, Gary Schells, Anna Leavitt (Sarasota County Government - UF/IFAS Extension)
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- **Center Lead:** Megan Rich (NASA DEVELOP at Georgia – Athens)
- **Modeling/Coding Guidance:** Dr. Kenton Ross (NASA DEVELOP Program Manager)

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PCA Analysis

Variable Loadings

